Table S1. Search strategy for this systematic review.

Database	Search date = 11/23/2020	Number of studies
Pubmed	**#1** ("diabetes mellitus"[MeSH Terms] OR "diabetes mellitus"[All Fields]) OR "diabetes"[All Fields]	742994
	#2 ("breath tests"[MeSH Terms] OR "breath"[All Fields]) OR ("exhalation"[MeSH Terms] OR "exhaled"[All Fields]) OR "expired gas"[All Fields] OR "expired air"[All Fields]	61429
	#3 (**#1** AND **#2**)	1427
	#4 limit **#**3 to humans	1143
	#5 limit #4 to English language	**1056**
Embase	**#1** 'diabetes':ab,ti OR 'diabetes mellitus':ab,ti	817631
	#2 'breath':ab,ti OR 'exhalation':ab,ti OR 'exhaled':ab,ti OR 'expired air':ab,ti OR 'expired gas':ab,ti	89470
	#3 **#1** AND **#2**	2398
	#4 limit **#**3 to humans	2200
	#5 limit #4 to English language	**2117**
Ovid MEDLINE	**#1** exp diabetes mellitus/	431737
	#2 (diabetes or "diabetes mellitus").ti,ab.	530068
	#3 (#1 or #2)	660386
	#4 exp breath test/	15065
	#5 (breath or exhaled or exhalation or (expired adj2 gas) or (expired adj2 air)).ti,ab.	55181
	#6 (#4 or #5)	58517
	#7 (#3 AND #6)	1132
	#8 limit #7 to humans	920
	#9 limit #8 to English language	**832**
Cochrane Library	**#1** MeSH descriptor: [Diabetes Mellitus] explode all trees	31368
	#2 (diabetes):ti,ab,kw OR (diabetes mellitus):ti,ab,kw	80884
	#3 (#1 OR #2)	83304
	#4 MeSH descriptor: [Breath Tests] explode all trees	1535
	#5 (breath):ti,ab,kw OR (exhaled):ti,ab,kw OR (exhalation):ti,ab,kw OR (expired gas):ti,ab,kw OR (expired air):ti,ab,kw	12706
	#6 (#4 OR #5)	12706
	#7 (#3 AND #6)	**361**
Science Citation Index Expanded	**#1** TOPIC: (diabetes) OR TOPIC: (diabetes mellitus)	808,330
	#2 TOPIC: (breath) OR TOPIC: (exhaled) OR TOPIC: (exhalation) OR TOPIC: (expired gas) OR TOPIC: (expired air)	91,049

	#3 (#1 AND #2)	1918
	#4 limit #3 to English language	**1859**

Table S2. Characteristics of included studies on diabetic breath analysis.

Author	Year	Country	Diabetes type	Number of patients/ controls	Breath condition control	Analytical techniques	Breath biomarkers	Sensitivity	Specificity	Accuracy	AUC
Kalidoss et al.[1]	2020	India	Indiscriminate	30/30	Fasting/2h post prandial	Sensor	Breathprint (Acetone(+))	66.66%	70.00%	68.33%	-
Kistenev et al.[2]	2020	Russia	Indiscriminate	11/11	Fasting	Terahertz spectroscopy	Acetone(+)	-	-	-	-
Tazi et al.[3]	2019	Indonesia	Indiscriminate	25/25	2h fasting	Sensor	Breathprint	-	-	-	-
[Δ]Kalidoss et al.[4]	2019	India	Indiscriminate	17/13	Fasting/2h post prandial	Sensor	Breathprint	70.05%	69.23%	69.64%	-
Trefz et al.[5]	2019	Germany	T1D	53/60	-	PTR-MS	Ethanol(+) Isopropanol(+) Dimethyl-sulfide(+) Isoprene(+) Pentanal(+)	-	-	-	-
Nevezhin et al.[6]	2019	Russia	T2D	22/14	No control	laser optical-acoustic IR spectroscopy	Breathprint	-	-	-	-
Yu et al.[7]	2019	China	Indiscriminate	16/16	Overnight fasting	GC-FID	Acetone(+) Isopropanol(+)	-	-	-	-
Som et al.[8]	2018	India	T2D(+HP)	37/51	Overnight fasting/ ^{13}C-glucose ingestion	IR	$^{13}CO_2$ * (-)	100.00%	96.90%	98.27%	0.993
Jeon et al.[9]	2018	Korea	T2D	12/13	2h	Sensor	Breathprint	-	-	-	-

					postprandial						
Tyas et al.[10]	2018	Indonesia	T2D	31/31	-	Laser photoacoustic spectrometer	Acetone(+)	-	-	-	-
Takemoto et al.[11]	2018	Japan	T2D	20/16	Overnight fasting/ ^{13}C-glucose ingestion	ND-IR	$^{13}CO_2$ * (-)	-	-	-	-
Yucel et al.[12]	2018	Turkey	PD	10/10	-	Sensor	Breathprint	-	-	-	-
Ghosh et al.[13],	2017	India	T2D	23/28	Overnight fasting/ ^{13}C-glucose ingestion	Laser-based integrated cavity output spectroscopy (ICOS)	$^{13}CO_2$ * (-) $C^{18}O_2$ * (+)	95.70%	**89.3% (95.8%)**	92.10%	0.93
Chien et al.[14]	2017	Japan	T1D (n=4) T2D (n=21)	25/55	No smoking in 1h; No alcohol drinking in 24h; 15min rest for gas exchange	Sensor	Acetone(+) Isopropanol(+)	-	-	-	-
Polanowska et al.[15]	2017	Poland	T2D(+chronic kidney disease)	10/10	-	GC-MS	Trimethylamine(+) Metanethiol(+) Dimethyl-sulfide(+)	-	-	-	-
Li W et al.[16]	2017	China	T2D	85/56	Overnight fasting/ Exercise restrained/	GC-MS	Acetone	62.40%	83.90%	70.9%	0.80
							Isopropanol	75.30%	85.70%	79.4%	0.86
							Acetone + isopropanol	71.80%	87.50%	78.0%	0.86

					10 min rest for gas exchange						
Jiang, et al.[17]	2016	China	T1D (n=22) T2D (n=312)	334/52	Overnight fasting/ postprandia	Cavity ringdown spectroscopy (CRDS)	Acetone(+)	-	-	74% for T1D	-
Ghosh et al.[18]	2016	India	T1D (n=14)	38/27	Overnight fasting/ ^{13}C-glucose ingestion	Laser-based ICOS	$C^{18}O_2$ * (+) T1D	92.90%	100%	96.40%	0.991
			T2D (n=24)				$C^{18}O_2$ * (-) T2D	-	-	-	-
Sun et al.[19]	2015	China	T2D	149/42	Overnight fasting/ 2h postprandial	GC-MS/ CDRS	Acetone(+)	-	-	-	-
Ghosh et al.[20]	2015	India	PD (n=39)	87/32	Overnight fasting/ glucose ingestion	Laser-based ICOS	$C^{18}O_2$ * (+)	94.90%	90.60%	92.3%	0.977
			T2D (n=48)					96%	92%	94.2%	0.957
Obermeier et al.[21]	2015	Germany	Indiscriminate	15/14	-	Sensor	Aldehydes(+)	-	-	-	-
▵Ghosh et al.[22]	2015	India	PD (n=38)	85/31	Overnight fasting/ ^{13}C-glucose ingestion	Laser-based ICOS	$^{13}CO_2$ * (-)	92.10%	100%	95.90%	0.963
			T2D (n=47)					95.70%	95.70%	95.30%	0.954
Rydosz[23]	2014	Poland	T1D	14/11	2h after glucose ingestion	LTCC-MS	Acetone(+)	-	-	-	-
Yan K et al.[24]	2014	China	T1D (n=1)	87/n>25	2h post	Sensor	Breathprint	91.51%	90.77%	91.14%	n.a

			T2D(n=86)		prandial						
Zhou M et al.[25]	2014	China	T2D	25/44	Overnight Fasting/ Exercise restrained	GC-MS	Acetone(+)	100%	97.7%	98.6%	n.a
Yan Y et al.[26]	2014	China	T2D	48/39	-	GC-MS	Acetone(+)	45.8%	87.2%	64.4%	0.679
							Isopropanol(+)	79.2%	92.3%	85.1%	0.876
							Toluene(+)	66.7%	82.1%	73.6%	0.737
							2,3,4-Trimethylhexane(+)	89.6%	94.9%	92.0%	0.91
							2,6,8-Trimethyldecane(+)	89.6%	94.9%	92.0%	0.949
							Tridecane(+)	89.6%	84.6%	87.4%	0.87
							Undecane(+)	89.6%	82.1%	86.2%	0.911
							m-Xylene(-)	**72.9% (69.2%)**	**69.2% (72.9%)**	71.3%	0.677
							Combination of 5 VOCs	97.9%	100%	98.9%	~1.00
^Δ^Ghosh et al.[27]	2014	India	PD (n=22)	47/18	Overnight fasting/ ^{13}C-glucose ingestion	RGA-MS	$^{13}CO_2$ * (-)	100%	94.40%	97.5%	0.99
			T2D (n=25)					100%	95.50%	97.8%	0.99
Goerl et al.[28]	2013	Germany	T1D (+end-stage renal disease)	10/20	-	GC-MS	Acetone(-)	-	-	-	-
Tanaka et al.[29]	2013	Japan	Indiscriminate	15/40	Overnight fasting/ ^{13}C-glucose ingestion	IR	$^{13}CO_2$ * (-)	100%	95%	96.4%	-
Tsuchiya et al.[30]	2012	Japan	T2D (+liver cirrhosis)	11/27(19 IGT)	Overnight fasting/	IR	$^{13}CO_2$ * (-)	81.8%	85.2%	84.2%	0.88

					^{13}C-glucose ingestion						
▵Yan K et al.[31]	2012	China	Indiscriminate	404/294	-	Sensor	Breathprint	91.43%	89.86%	90.65%	-
Guo et al.[32]	2010	China	Indiscriminate	117/108	-	Sensor	Breathprint	87.67%	86.87%	-	-
Greiter et al.[33]	2010	Germany	T2D	21/26	Overnight fasting/ Exercise restrained	PTR-MS	Acetone (+)	76.2%	80.8%	78.7%	-
							Combination of 8 endogenous VOCs	90.5%	92.3%	91.5%	-
▵Guo et al.[34]	2010	China	Indiscriminate	90/108	-	Sensor	Breathprint	93.52%	92.66%	92.93%	-
Wang C et al.[35]	2010	China	T1D (n=34) T2D (n=10)	44/15	-	CRDS	Acetone(+)	-	-	-	-
Deng et al.[36]	2004	China	T2D	15/15	-	GC-MS	Acetone(+)	-	-	-	-
Yatscoff et al.[37]	2003	Canada	T2D	52/50 (+2 undefined)	Overnight fasting/ ^{13}C-glucose ingestion	IR-MS	$^{13}CO_2$ * (-)	73.0%	92.0%	82.5%	-
Zhang et al.[38]	2000	China	Indiscriminate	18/14	preprandial/ 2h postprandial	Sensor/ GC	Breathprint/ Acetone(+)	-	-	-	-
Paredi et al.[39]	1999	UK	T1D (n=8) T2D (n=16)	24/37	-	Sensor	CO(+)	-	-	-	-
Wang P et al.[40]	1997	China	T1D (Idiopathic)	18/14	Preprandial/ 2h postprandial	Sensor	Breathprint	-	-	-	-
Lefebvre et al.[41]	1975	Belgium	T1D (n=5) CD (n=7)	12/11	Overnight fasting/	MS	$^{13}CO_2$ * (-)	-	-	-	-

					^{13}C-glucose ingestion						
Tassopoulos et al.[42]	1969	UK	Indiscriminate	251/45	Overnight fasting	GC-FID	Acetone(+)	-	-	-	-
Rooth, et al.[43]	1966	Sweden	T1D (n=49) T2D (n=20) PD (n=40) CD (n=42)	151/67	-	GC-FID	Acetone(+)	-	-	-	-
Stewart et al.[44]	1964	USA	T1D(n=33) T2D(n=96)	129/40	Overnight Fasting	GC-FID/ IR	Acetone(+)	-	-	-	-

*Biomarkers from exogenous test meal.

▵Same population may be used, these studies were excluded in meta-analysis.

(+) Means increasing in diabetic exhaled breath, (-) means decreasing in diabetic exhaled breath.

Data in **bold gray font** were corrected according the available data provided in the literatures, and (x) were data provided in original text.

Table S3. Detailed population information of the included studies. The FPG units were all converted to mg/dL, while all HbA1c were converted to DCCT-HbA1c (%).

Studies	Number of participates in each group	Age	Gender (male/female)	BMI	FPG mg/dL	HbA1c/%
Kalidoss et al., 2020	30 DM	37 ± 9	13/17	25.7 ± 3.2	123.2 ± 21.9	7.00 ± 2.97
	30 ND	44 ± 9	17/13	26.4 ± 3.1	103.4 ± 21.8	5.86 ± 2.98
Kistenev et al., 2020	11 DM	-	-	-	-	-
	11 ND					
△Kalidoss et al., 2019	17 DM	35±3	4/9	25.7±3.2	126.52±22.14	7.12±2.60
	13 ND	50±11	11/6	26.4±3.1	96.07±22.42	5.70±2.89
Trefz et al., 2019	53 T1D	12.36±3.15	32/21	0.35±0.69	-	8.60±(-)
	60 ND	13.67±2.80	28/32	0.06±0.82		-
Tazi et al.,2019	25 DM	-	-	-	-	-
	25 ND					
Nevezhin et al., 2019	22 T2D	-	-	-	-	-
	14 ND					
Yu et al., 2019	16 DM	-	-	-	-	-
	16 ND					
Jeon et al., 2018	12 T2D	55.45±16.86	6/6	24.0±3.7	-	7.2±1.7
	13 ND	34.92±11.26	4/9	21.7±3.0		5.2±0.3
Som et al.,2018	37 T2D (+Helicobacter pylori)	50 ± 9.5	73/58	-	159±11.4	10.7±1.1
	51 ND (+Helicobacter pylori)	46 ± 7			101±6.4	5.31±0.27
	43 ND	47 ± 10			97±10.1	5.24±0.19
Yucel et al., 2018	10 PD	-	-	-	100-125	-
	10 ND				-	
Takemoto et al., 2018	20 T2D	58.4±14.4	15/5	24.6 ± 4.0	-	9.2±2.2
	16 ND	21.5±1.0	7/9	20.8 ± 2.7		-
Tyas et al., 2018	31 T2D	25-80	-	-	-	-
	31 ND					
Polanowska et al.,2017	10 T2D (+chronic kidney disease)	49.4±6.4	4/6	-	-	-
	10 ND	38.1±2.7	4/6			
Chien et al., 2017	4 T1D	44.5 ± 17.2	9/16	-	-	-
	21 T2D	64.5 ± 10.5				
	55 ND	30.8 ± 10.9	41/14			
Ghosh et al. ,2017	28 T2D	36.1±8.5	14/9	24.8±1.7	119.6±13.4	8.6±1.1
	23 ND	33.2±10.1	17/11	23.7±1.9	237.2±49.2	5.3±0.3
Li W et al., 2017	85 T2D	20-85	41/44	-	270.0±121.86	10.56±2.60
	56 ND	21-71	21/32		-	-
Jiang et al.,	22 T1D/	28.1±13.4	8/14	20.6±2.5	203.3±53.9	n.a

2016	312 T2D	57.0±12.9	178/134	26.5±3.8	152.1±41.0	
	52 ND	27.0 ±4.8	22/30	22.1±3.1	91.8±7.3	
Ghosh et al., 2016	14 T1D	21.7±13.8	8/6	19.4±2.9	272.6±107.6	11.3±2.9
	24 T2D	36.2±11.7	15/9	24.6±2.3	132.08±12.2	7.5±0.5
	27 ND	31.7±8.9	16/11	23.9±2.4	87.7±9.8	5.2±0.2
Sun et al., 2015	149 T2D/	57.2±13.4	67/82	26.2±3.8	-	-
	42 ND	27.8±6.3	23/19	22.7±3.9		
Ghosh et al., 2015	48 T2D	38.6 6±9.7	19/29	23.88±2	90.19±6.4	5.27±0.2
	39 PD	34.64±6.5	14/25	24.17±2.4	107.05±5.1	6.11±0.2
	32 ND	35.94±7.15	13/19	24.28±2.8	140.42±7.7	7.65±0.5
Obermeier et al., 2015	15 DM/	-	7/8	-	-	-
	14 ND		7/7			
△Ghosh et al., 2015	47 T2D	38.63±9.8/	31/16	23.93±2	196±17.6	7.64±0.5
	38 PD	34.55±6.6/	26/12	24.21±2.5	107.2±9.1	6.07±0.2
	31 ND	35.48±6.8	18/13	24.43±2.7	90.1±12.5	5.26±0.2
△Ghosh et al., 2014	25 T2D	50.08±15.4	15/10	23.36 ± 1.2	221.52±80.3	9.63±1.6
	22 PD	51.45±15.3	12/10	23.83 ± 0.7	110.64±10.8	6.11±0.2
	18 ND	41.72±11.5	11/7	23.37 ± 0.8	98.61±11.8	5.28±0.2
Yan K et.al., 2014	1 T1D/86 T2D	39-91	39 / 48	-	-	-
	n>25 ND	n.a	-			
Zhou M et al., 2014	25 T2D	42.7±2.9	12/13	-	-	-
	44 ND	29.3±1.8	25/19			
Yan Y et al., 2014	48 T2D	14-85	22/26	-	-	-
	39 ND	21-71	15/24			
Rydosz, 2014	14 T1D	32.09±6.01	6/5	-	-	-
	11 ND	28.21±9.74	6/8			
Tanaka et al., 2013	15 DM	-	8/7	-	-	-
	40 ND		19/21			
Goerl et al., 2013	10 T1D(+end-stage renal disease)	73.3±6.5	5/5	-	-	-
	20 ND(+end-stage renal disease)	60.7±14.7	11/9			
Tsuchiya et al., 2012	11 DM (+liver cirrhosis) 19 IGT(+liver cirrhosis)	67.5±10.5	23/15	23.1±3.0	93±10.3	5.1±0.4
	8 ND(+liver cirrhosis)					
△Yan K et al., 2012	404 DM	-	-	-	-	-
	294 ND					
Guo et al., 2010	117 DM	32-70	65/52	-	-	-
	108 ND	23-60	58/50			

△Guo et al., 2010	90 DM	25-77	49/41	-	-	-
	108 ND	23-60	58/50			
Greiter et al., 2010	21 T2D	66.1±8.3/	-	28.9±4.9	-	7.7±1.4
	26 ND	40-59		-		-
Wang C et al., 2009	34 T1D	46.6±19.5	14/20	-	-	7.8±1.9(n=20)
	10 T2D	54.5±12.2	3/6(a unknown)			8.8±1.9(n=3)
	15 ND	25–45	-			-
Deng et al., 2004	15 T2D	54.1±(-)	7/8	-	-	-
	15 ND	47.6±(-)	7/8			
Yatscoff et al., 2003	54 T2D	54±11	-	31.6±5.5	-	-
	50 ND	44±14		28.8±5.2		
Zhang et al. , 2000	18 DM	-	-	-	-	-
	14 ND					
Paredi et al., 1999	8 T1D	50 ± 8	4/4	-	-	7.5±0.3
	16 T2D	63 ± 8	5/11			4.9±0.4
	37 ND	33 ± 3	20/17			-
Wang et al., 1997	18 DM	-	-	-	-	-
	14 ND					
Lefebvre et al., 1974	5 T1D	22.6±9.2	5/0	19.3 ± 1.3	-	-
	7 CD	35.3±13.3	2/5	32.4 ± 6.5		
	11 ND	25.6±4.8	9/2	21.7 ± 2.8		
Tassopoulos et al., 1969	251 DM	-	-	-	-	-
	45 ND		21/24			
Rooth et al., 1966	49 T1D	-	-	-	-	-
	20 T2D					
	40 PD					
	42 CD					
	67 ND					
Stewart et al., 1964	33 T1D	-	-	-	-	-
	96 T2D					
	40 ND					

Table S4. The QUADAS-2 referenced to the *Hanna, et. al.*'s version.

Applicability concern	
Patient selection	
	Were sampled patients representative of the intended population?
	Did the study include both positive (benign conditions) as well as healthy controls?
	Did the study avoid inappropriate exclusions?
Index test	
	Was the index test and interpretation of data performed in standardised and reproducible fashion?
	Was validation of results performed (internal or external)?
Reference standard	
	Is the reference standard likely to correctly classify the target condition?
Flow and timing	
	Was there an appropriate interval between index test and reference standard?
	#Was diet control performed prior to index test and reference standard ?
	Did all patients receive the same reference standard?
	Were all patients included in the analysis?
Applicability concern	
Patient selection	Are there concerns that the included patients and setting do not match the review question?
Index test	Are there concerns that the authors have not demonstrated suitable reproducibility and sensitivity of the chosen index test?
Reference standard	Are there concerns that the target condition as defined by the reference standard does not match the question?

The signalling question further modified

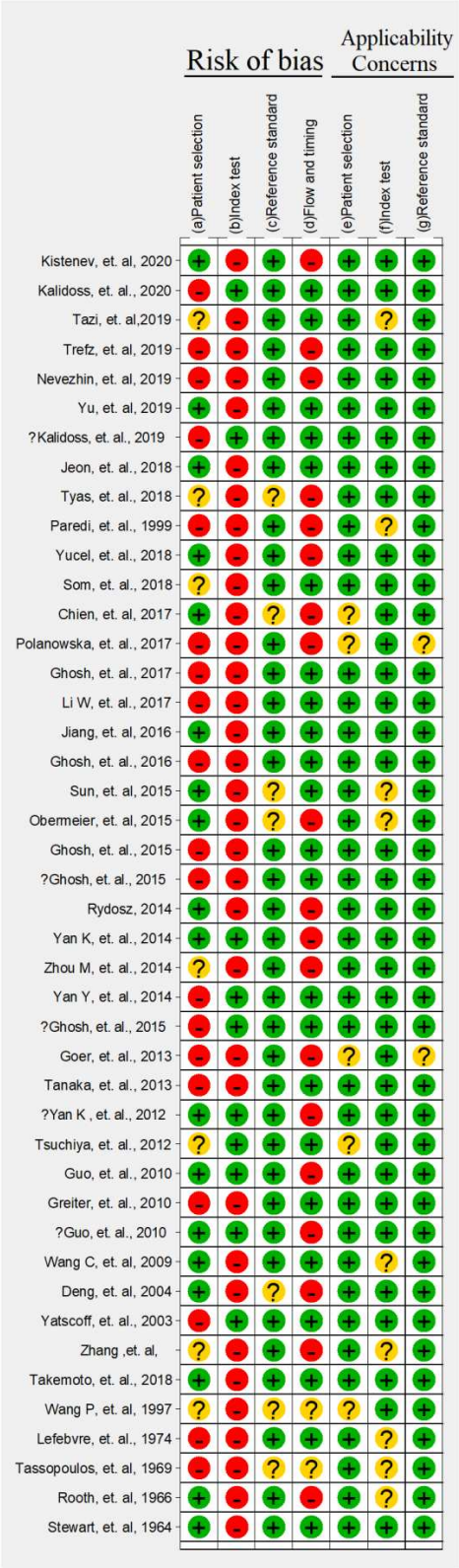


Figure S1. The summary of risk of bias and applicability concern.

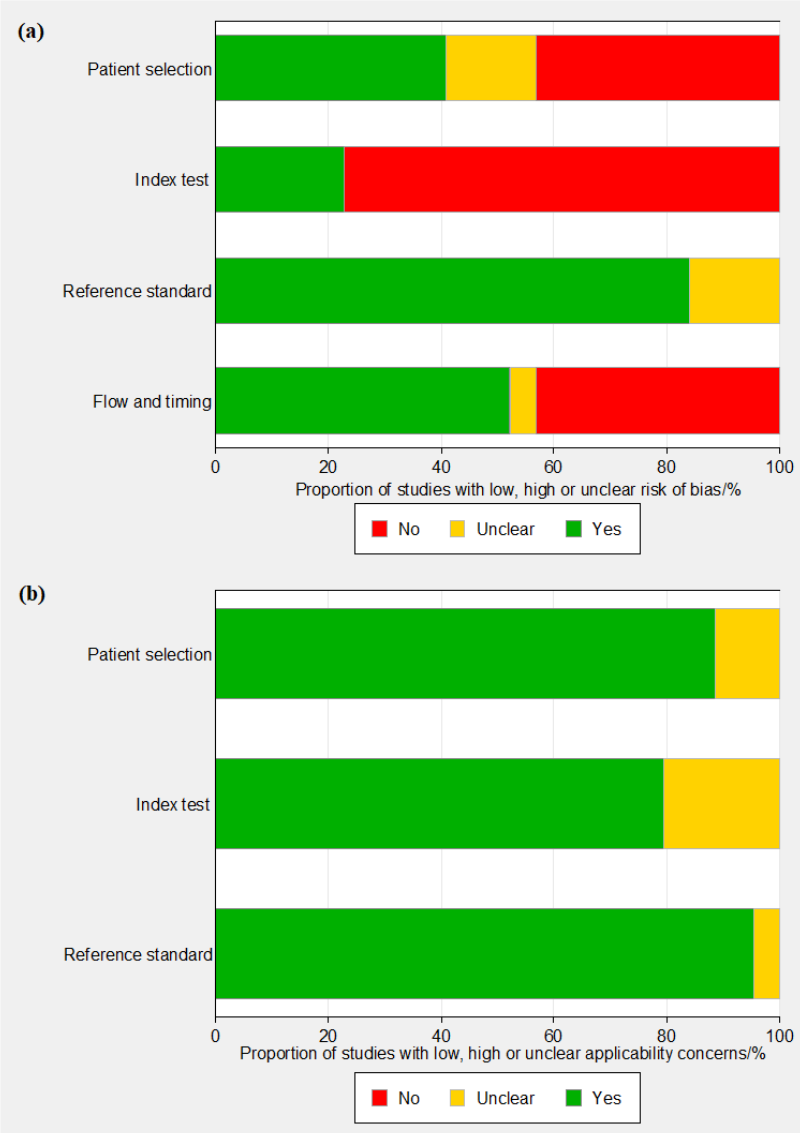


Figure S2. Risk of bias and applicability concern using QUADAS-2.

Table S5. The STARD assessment of each study.

Studies	Title	Abs.	Intro.		Methods																	Results								Discussion		Other information			Total
	1	2	3	4	5	6	7	8	9	10_a	10_b	11	12_a	12_b	13_a	13_b	14	15	16	17	18	19	20	21_a	21_b	22	23	24	25	26	27	28	29	30	
Kalidoss, 2020	1	0	1	1	1	1	0	1	1	1	1	1	0	1	1	1	1	0	0	0	0	0	1	1	0	1	1	0	0	0	1	0	0	1	20
Kistenev, 2020	0	0	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	6
Trefz , 2019	0	0	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	1	1	0	1	11
Yu, 2019	0	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	9
Nevezhin, 2019	0	0	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6
Tazi, 2019	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	3
△Kalidoss, 2019	1	0	1	1	1	1	0	1	0	1	1	1	0	1	1	0	1	0	0	0	0	0	1	0	0	1	1	0	0	1	1	0	0	1	18
Som, 2018	1	0	0	1	1	1	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	1	1	0	0	0	0	0	0	1	1	1	0	1	17
Jeon, 2018	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	1	0	1	7
Tyas, 2018	0	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	5
Yucel, 2018	0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	5
Takemoto, 2018	0	0	1	0	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	9
Chien et. al, 2017	0	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	1	1	0	1	11
Ghosh,2017	1	0	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1	0	1	18
Li W, 2017	1	0	1	1	1	1	0	1	1	1	1	0	0	1	1	0	1	0	0	0	0	0	1	1	0	1	0	1	0	0	1	0	0	1	18
Polanowska,2017	0	0	1	0	1	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	1	8
Jiang, 2016	0	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	10
Ghosh, 2016	0	0	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	1	1	0	0	0	1	0	0	1	1	0	1	17
Ghosh, 2015	0	0	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	1	1	0	1	0	1	0	0	1	1	0	1	18
Sun, 2015	0	0	1	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	1	1	0	1	11
Obermeier , 2015	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	2
△Ghosh, 2015	0	0	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	1	1	0	1	0	0	0	0	1	1	0	1	17

Yan K, 2014	1	0	1	1	1	0	0	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	1	1	0	1	0	0	0	0	1	0	0	1	14
Zhou M, 2014	1	0	1	1	1	0	0	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	13
Yan Y, 2014	1	0	1	1	1	1	0	1	1	1	1	0	0	1	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	1	15
Rydosz, 2014	0	0	1	0	1	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	1	9
[▵]Ghosh, 2014	1	0	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	1	1	0	0	1	1	0	0	1	1	0	1	19
Tanaka, 2013	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	1	0	0	1	16
Goerl, 2013	0	0	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	5
Tsuchiya, 2012	1	1	1	1	1	0	0	1	1	1	1	0	1	1	1	0	1	0	0	0	0	0	1	1	0	1	0	1	0	0	1	0	0	0	18
[△]Yan K, 2012	0	0	1	1	1	0	0	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	1	11
Guo, 2010	0	0	0	1	1	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	1	10
Greiter, 2010	1	1	1	1	1	1	1	0	0	1	1	0	0	1	1	0	1	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	1	16
[△]Guo, 2010	0	0	1	1	1	0	0	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	1	0	0	1	13
Wang C, 2009	0	0	1	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0	1	0	0	1	14
Deng, 2004	0	0	1	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	1	9
Yatscoff, 2003	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	1	1	0	0	0	1	0	0	0	18
Zhang, 2000	0	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	8
Paredi, 1999	0	0	1	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	7
Wang P, 1997	0	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	8
Lefebvre, 1974	0	0	0	0	1	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	1	8
Tassopoulos,1969	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	4
Rooth, 1966	0	0	0	0	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	1	8
Stewart, 1964	0	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0	1	0	0	0	9

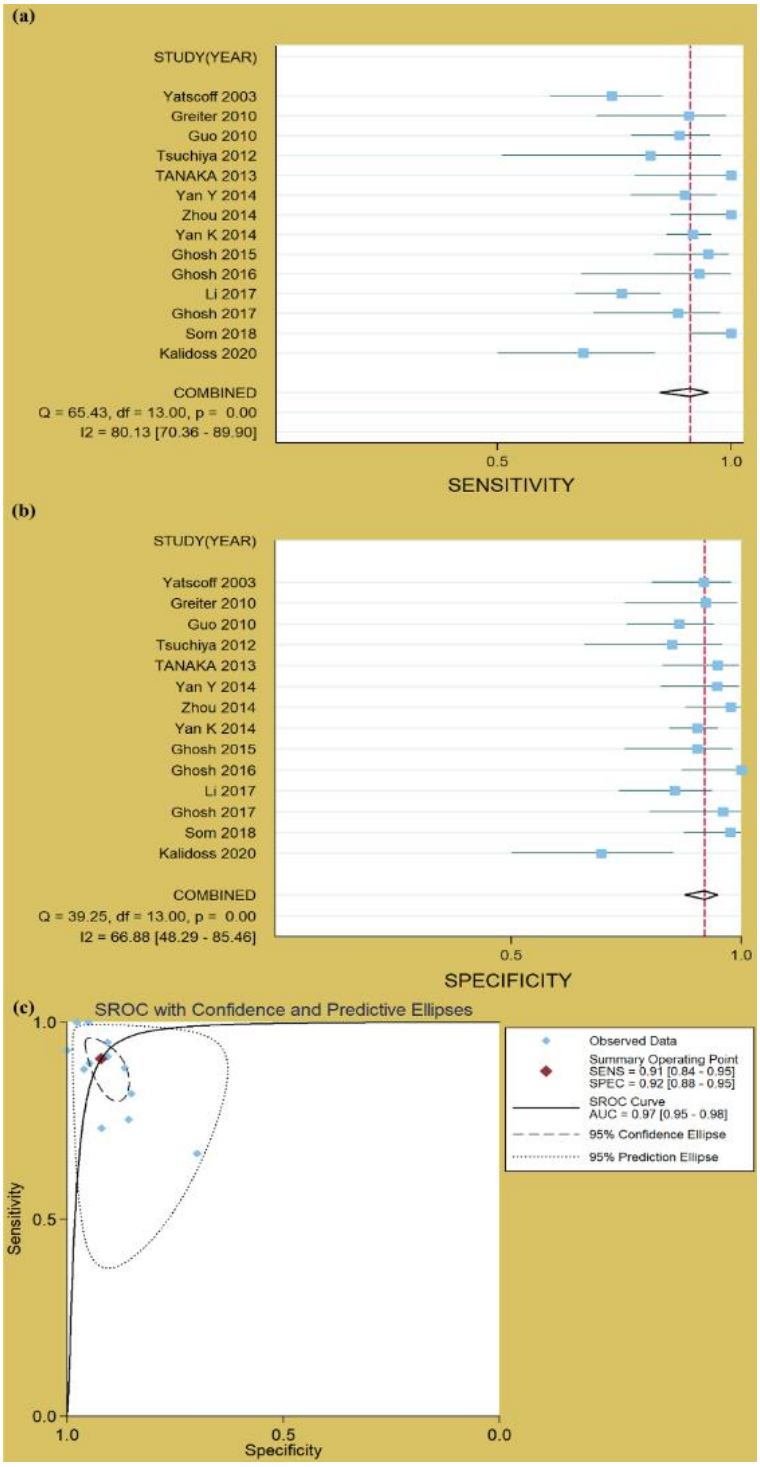


Figure S3. The forest plot of (a) sensitivity, (b) specificity and (c) the summary receiver operating characteristic curve for overall breath tests.

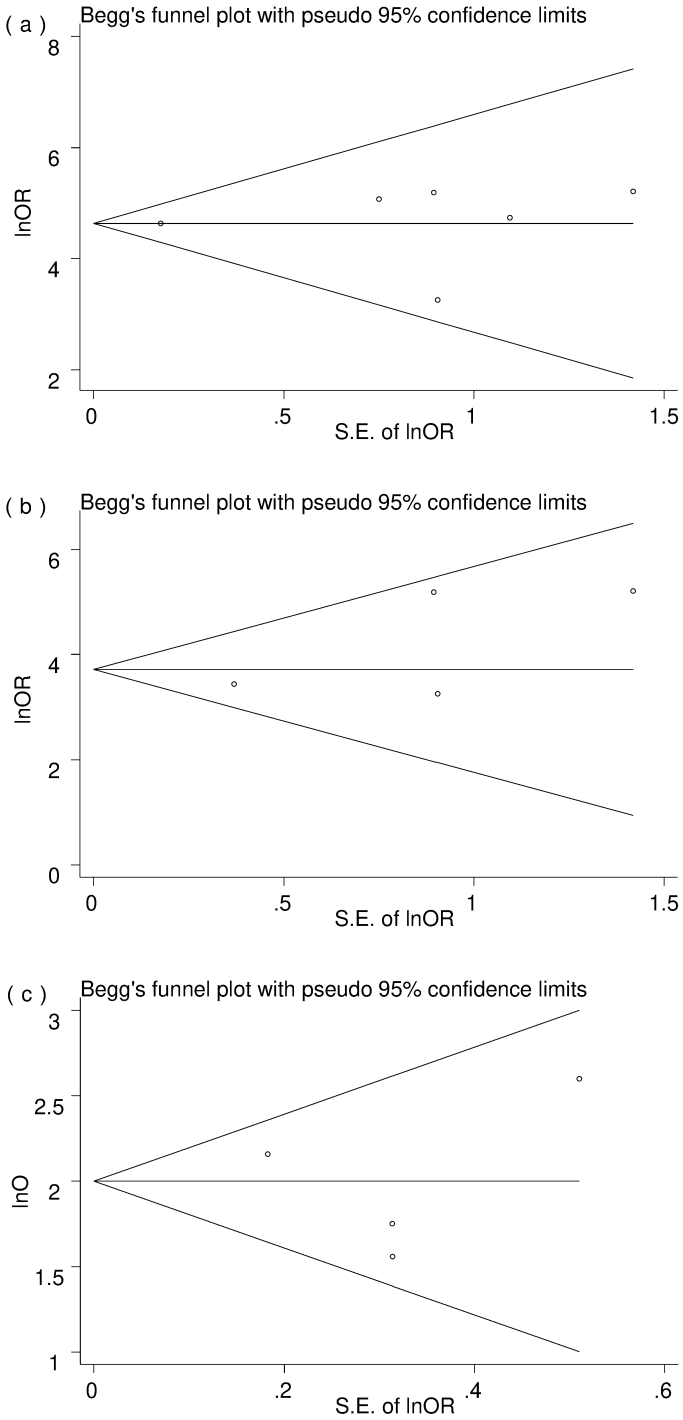


Figure S4. The funnel plot of the publication bias. (a) T2D group, (b) isotopic CO_2, (c) acetone.

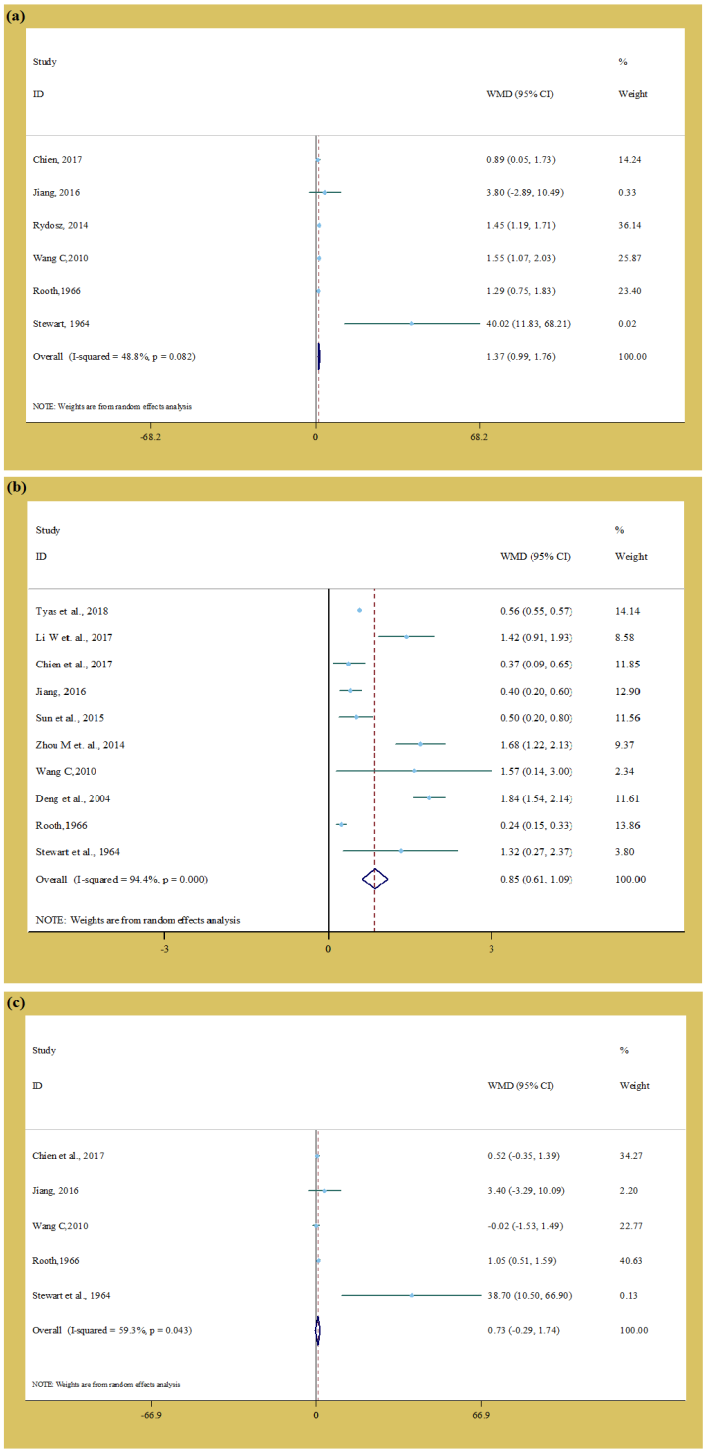


Figure S5. A random effect model was applied, and the effect sizes for the comparison of breath acetone concentrations in (a)T1D vs ND, (b) T2D vs ND, (c) T1D vs T2D were presented. The comparison of T1D vs ND, T2D vs ND, and T1D vs T2D exhibited WMD of 1.374 (95%CI 0.986-1.762, z=6.94, p=0.000), 0.845 (95%CI 0.605~1.085, z=6.91, p=0.000), and 0.725 (95%CI -0.291~1.741, z=1.40, p=0.162), respectively.

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